National University of Lesotho

Department of Mathematics and Computer Science

M3503 – Ordinary Differential Equations

Tutorial 1 • First-Order Equations

(August 18, 2026)

Instructions

Attempt *all* questions. Classify each equation in Question 1 before solving (**separable**, **linear**, **exact**, **homogeneous**, requiring an **integrating factor**, or amenable to a **substitution**), then choose the appropriate method. Show all working.

1. Core exercises

1. Solve the following.

(a) $e^x y\,\dfrac{dy}{dx} = e^{-y} + e^{-2x-y}$

(b) $x^2\dfrac{dy}{dx} = y - xy, \qquad y(-1) = -1$

(c) $y\ln x\,\dfrac{dx}{dy} = \left(\dfrac{y+1}{x}\right)^2$

(d) $\csc y\,dx + \sec^2 x\,dy = 0$

(e) $\sqrt{1-y^2}\,dx - \sqrt{1-x^2}\,dy = 0, \qquad y(0) = \dfrac{\sqrt{3}}{2}$

(f) $x^2y' + x(x+2)y = e^x$

(g) $y' + 3x^2y = x^2$

(h) $x\dfrac{dy}{dx} - y = x^2\sin x$

(i) $(5x+4y)\,dx + (4x - 8y^3)\,dy = 0$

(j) $(\sin y - y\sin x)\,dx + (\cos x + x\cos y - y)\,dy = 0$

(k) $\left(1 + \ln x + \dfrac{y}{x}\right)dx = (1 - \ln x)\,dy$

(l) $(2y^2 + 3x)\,dx + 2xy\,dy = 0$

(m) $\cos x\,dx + \left(1 + \dfrac{2}{y}\right)\sin x\,dy = 0$

(n) $(10 - 6y - e^{-3x})\,dx - 2\,dy = 0$

(o) $(3x^2 - y^2)\,dx + (xy - x^3y^{-1})\,dy = 0$

(p) $\dfrac{dy}{dx} = \dfrac{y^2 + x\sqrt{x^2+y^2}}{xy}$

(q) $\dfrac{dy}{dx} = (x - y + 5)^2$

(r) $\dfrac{dy}{dx} = \sin(x - y)$

2. According to **Newton's law of cooling**, if an object at temperature T is immersed in a medium of constant temperature M, the rate of change of T is proportional to $M - T$:

$$\frac{dT}{dt} = k(M - T).$$

A thermometer reading 100°F is placed in a medium held at 70°F. After 6 minutes it reads 80°F. What does it read after 20 minutes?

3. Find an integrating factor of the form $x^n y^m$ that makes each equation exact, then solve it.

(a) $(2y^2 - 6xy)\,dx + (3xy - 4x^2)\,dy = 0$

(b) $(12 + 5xy)\,dx + (6xy^{-1} + 3x^2)\,dy = 0$

2. Clever substitutions

4. Reduce each to a separable or linear equation by a suitable substitution, then solve.

(a) $\dfrac{dy}{dx} = \dfrac{1}{x+y}$ (try $u = x + y$)

(b) $\dfrac{dy}{dx} = \dfrac{x+y-1}{x+y+1}$

(c) $\dfrac{dy}{dx} = \dfrac{y-x+1}{y-x+5}$

(d) $(2x + 3y - 1)\,dx + (2x + 3y + 2)\,dy = 0$

(e) $x\,y' = y + x\,e^{y/x}$ (homogeneous)

(f) $\dfrac{dy}{dx} = \dfrac{2x+y}{x-2y}$

3. Exactness and integrating factors

5. Determine whether each equation is exact; if not, find an integrating factor depending on x *alone* or y *alone*, then solve.

(a) $(3xy + y^2)\,dx + (x^2 + xy)\,dy = 0$

(b) $\big(y^2 + 2xy\big)\,dx - x^2\,dy = 0$

(c) $\big(2xy^2 - 3y^3\big)\,dx + \big(7 - 3xy^2\big)\,dy = 0$

4. Modelling with first-order equations

Applications

Set up the governing equation, identify the type, solve it, and interpret the answer in the original units. Note the interval on which your model is physically meaningful.

6. Population (logistic). A population $P(t)$ obeys $\dfrac{dP}{dt} = P(a - bP)$ with $a = 0.1\,\text{yr}^{-1}$, $b = 10^{-4}\,\text{yr}^{-1}$, and $P(0) = 500$. Solve for $P(t)$ and find the carrying capacity.

7. Falling body with drag. An object of mass m falls under gravity with air resistance proportional to velocity: $m\dfrac{dv}{dt} = mg - kv$, $v(0) = 0$. Find $v(t)$ and the terminal velocity.

End of Tutorial 1.